

Document Title: HGV Eco Driving Training Standard

Document Number: HGV Eco driving Standard-2024

Version: 1.0

Effective Date: 2024-09-30

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Approval Authority: Department of Transport

Reviewed By: James Kearney (CILT on behalf of Department of Transport)

Next Review Date: 2025-09-30

Change History:

Version	Date	Description	Author	Approved By
1.0	2024-09-30	Initial release	CILT	Department of Transport

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Document Control

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Standard Requirements which will be subject to Audit

This Standard contains the essential elements required to meet the approval threshold for the Department of Transport HGV Eco Driving Standard.¹ It also contains best practice examples and suggestions which may aid training providers in delivering successful training interventions. The core standards which will be assessed as part of initial approval and ongoing auditing are shared in the document with the notation ‘*’ and listed within the appendix called Auditable Standards. Where a heading is marked as part of the core standard, the examples contained within the subheadings will form part of the assessment criteria, but they are not necessarily all required to meet the standard unless marked with an ‘*’ individually.

¹ It should be noted that the Department of Transport HGV Eco Driving Standard is a voluntary standard, awarded to those training providers who decide to apply and who meet the requirements set out in this document.

1. Scope

This standard specifies requirements for HGV Eco driving coaching programmes aimed at improving driving efficiency and reducing environmental impact through reducing fuel consumption, energy management & maintenance efficiencies. It covers aspects of coaching delivery, safety protocols, performance measurement, follow-up support, driver training and coaching, assessment and approval.

2. Terms and Definitions

- **Assessing Body:** an organisation appointed by the Department of Transport to devise oversee and update as needed standards applicable to Eco Driving Training in Ireland and to approve Eco Driving Training Providers, Trainers and Master Trainers. The assessing body will also maintain a panel of Subject Matter Experts (SME's) approved by the Department of Transport for the management and oversight of the standard.
- **Approved HGV Eco Driving Training Provider:** an organisation or person approved by the assessing body to offer and conduct eco driving training programmes to the public in line with these standards. Where the person is not an Approved HGV Eco Driving Master Trainer this person may not deliver in cab coaching or train an in-house coach.
- **Approved HGV Eco Driving Master Trainer:** a person who is a recognised ADI or has a minimum of 3 years licensed experience in the category of vehicle. They are a trainer approved by the assessing body in relation to these standards to deliver train the trainer training for in house coaches and to the public via an approved eco driving training provider.
- **Approved Driving Instructor (ADI):** A person who is currently registered with the RSA as an ADI for the relevant vehicle category and approved to teach learner drivers.
- **Coaching:** A developmental process whereby an in-house coach supports an appropriately licensed colleague in achieving a specific personal or professional goal by providing training and guidance.
- **Eco-Will:** Widespread Implementation for Learner Drivers and Licensed Drivers short duration eco-driving training programs for licensed drivers in 13 European countries. Information available at <https://www.cieca.eu/project/33>.
- **In-House Coach:** A non-ADI individual trained in line with these standards by an Approved HGV Eco Driving Master Trainer to deliver the HGV Eco driving coaching programme for professional drivers within their own organisation. In house coaches will also have completed a recognised train the trainer course.
- **Trip Computer:** A vehicle's built-in device that provides data on fuel consumption, distance travelled, and other performance metrics.

- **TREATISE:** EC project operating from January 2005 to June 2007 that provides free training on sustainable transport subjects, primarily for energy and transport professionals. [TREATISE Cleaner Fuels & Vehicles](#)
- **SAFED: Safe and Fuel-Efficient Driving (SAFED)** is a theory and practical driver training programme that aims to re-educate drivers in safer driving, modern fuel-efficient driving techniques and to reduce a driver's risk of being involved in a collision. [SAFED Final Report \(europa.eu\)](#)
- **Subject-Matter Expert (SME)** is someone who has extensive knowledge in Driver training and Eco-driving. This expertise is typically demonstrated through their degree, licensure, and years of professional experience, SMEs are proposed by the Assessing Body and approved by the Department of Transport.

3. Assumptions

3.1. Driver Competency

- The standards outlined here assume that all participants are already experienced competent & licensed HGV drivers. No basic driving instruction will be included within the eco driving standards.
- Where HGV Eco-Driving training is being delivered concurrently with new Driver Training, all in-cab instruction must be in compliance with established guidance of the Road Safety Authority (RSA). Only an ADI is allowed in the cab with a driver that does not hold a full license for the vehicle.

4. Safety Protocols

4.1. Where relevant, an ADI instructor must communicate to the driver that their presence in the cab is the role of HGV Eco-Driving trainer and the driver should not assume any advice offered is an instruction. The driver remains fully responsible for the vehicle and compliance with the rules of the road regardless of any advice offered by the trainer or coach.

*5. Coaching Delivery

5.1. HGV Eco-Driving Training Requirements

*Classroom based instruction should be led by an Approved HGV Eco Driving Master Trainer OR an Approved HGV Eco Driving Training Provider OR In-House Coach. The instructor must have appropriate knowledge/expertise specific to the content they are delivering, including, but not limited to:

- **Eco Will**
- **TREATISE**
- **SAFED**

Or

the assessing body will consider other qualifications that meet a similar standard on an individual basis.

- *In addition to familiarity with the criteria of this standard, all classroom based trainers will have completed at a minimum Train the Trainer (level 6) or other similar preparation to deliver educational courseware.

In cab training or coaching to the public can only be delivered by an Approved HGV Eco Driving Master Trainer.

Where a company chooses to use their own in-house coach for the dissemination of the training the following must be followed:

- All in house coaches must be trained by an Approved HGV Eco Driving Master Trainer.
- In House coach must be fully licensed for the vehicle in which they are coaching with a minimum of three (3) years' experience in the category of vehicle in which they are coaching.
- In house coach must remind the vehicle driver that the driver remains fully responsible for the vehicle and compliance with the rules of the road regardless of any advice offered by the coach.

5.2. *Delivery Method

- The training content will be comprised of various modules both theory and practical using face-to-face delivery. All in-cab training must be delivered in person.
- *HGV Eco Driving theory training will be undertaken for a minimum of 2 hours before any in cab sessions.
- *Training in cab on-road driver observation, advise and measurement.
- *Maximum number of personnel in the vehicle should not exceed 3 - the Trainer, Driver, Observer
- The number of seatbelts should be sufficient to cover everyone in the cab
- Classroom based training may include planners and other non-driving support staff.
- *Results must be evidenced using the vehicle's trip computer or other appropriate data capture mechanism.
- Approved HGV Eco Driving Training Providers should encourage the senior management to be present on Day 1 to emphasize the commitment of the organisation and the importance of the driver's efforts.

5.3.* Learning Outcomes - After completing this programme, the driver will:

- Understand HGV Eco driving principles and appreciate the environmental and economic benefits.
- Know practical techniques to improve fuel consumption and achieve measurable reductions in fuel usage.
- Be familiar with the range of alternative power propulsion technologies and fuels on the market, EV/Hybrid, Hydrogen, HVO - CNG/biomethane.

5.4. *Indicative Programme Content-Theory Module

- Introduction to Eco-Driving
- Driver Awareness and Augmented Reality Technologies

- Fuel Efficiency Techniques
- Vehicle Knowledge
- Power / Torque/HP/BHP/kWh/Calorific Value
- Specific Fuel Consumption: Partial Load – Full Load
- Propulsion Types EV/Hybrid/Hydrogen /Diesel/CNG/Biogas
- Regeneration (electric and hybrid vehicles)
- The Need for Exhaust Aftertreatment Systems
- Maintenance Intervals
- Manual & Automatic/AMT Gearbox:
 - The modern automatic gearbox (different types: DSG/AMT, Fully Auto (torque converter automatic, variomatic); selectable driving mode
 - Manual shifting
- Tyres:
 - Tyre pressure, direct or indirect Tyre Pressure Monitoring System (TPMS)
 - Using winter tyres in summer
 - Low rolling resistance tyres, Eco Tyres and EV tyres
- Temperature Inside the Vehicle:
 - Air conditioner/HVAC use
 - Open windows
- Electricity Consumers Increase Fuel Consumption
- Driving Techniques
- Plan Your Route
- Starting and Driving
- Speed, Fuel Consumption and Speed
- Accelerating Correctly
- Using Momentum
- Driving Style – Most Important Tips:
 - Plan optimally (route planning, traffic jams, mountain route)
 - Do not wait for the engine to warm up
 - Move off in low gear
 - Only roll briefly, then shift gears
 - Use short acceleration phase utilising the green range on the tachometer/Rev Counter up to the target speed
 - Optimise engine speed and gear changes
 - Use as high a gear as possible in the part-load range
 - Drive uphill in the maximum torque range (do not downshift every half-gear)
 - Use the exhaust brake/retarder for optimum efficiency
 - Use momentum (road dips)
 - Take your foot off the accelerator in good time
 - Read the road ahead to maintain a safe well-planned speed
- Additional Eco-Driving Measures
- Personal Factors Affecting Driving
- Review and Q&A

5.5 *Practical Module

*First Drive: In-Vehicle Driver Evaluation (Minimum 25km driving)

- **Objective:** Assess current driving skills and identify areas for improvement.
- **Activities:**
 - Initial driving assessment with an Approved HGV Eco Driving Master Trainer OR In-House Coach
 - Real-time feedback and coaching during the drive.
 - Use of telematics to monitor driving behaviours (e.g. RIBAS system).

*Fuel Efficiency and Safety Checks

- **Objective:** Teach drivers how to perform routine checks to ensure vehicle efficiency and safety.
- **Activities:**
 - Checking tyre pressure and condition.
 - Inspecting fluid levels (oil, coolant, brake fluid).
 - Ensuring proper functioning of lights and indicators.
 - Demonstrating the impact of proper maintenance on fuel efficiency.

*Second Drive: Driver Coaching (25km Minimum)

- **Objective:** Provide targeted coaching to improve driving skills and habits.
- **Activities:**
 - One-on-one coaching sessions.
 - Focus on specific areas such as smooth acceleration, efficient braking, and optimal gear usage.
 - Use of simulation tools for practice if available.
 - Practicing efficient driving techniques (e.g., using momentum, avoiding unnecessary idling).
 - Driving in different conditions (urban, highway, hilly terrain).
 - Use of telematics to monitor driving behaviours

*Final Evaluation

- **Objective:**
 - Assess the improvement in driving skills and knowledge.
 - Recognize the completion of the training program.
- **Activities:**
 - Issuance of a certificate of completion.
 - Highlighting key achievements and areas of excellence.

6. Performance Measurement

6.1. *Assessment Strategy

Drivers participating in the programme are to be assessed using a range of methods (below) to test driver competence vs the learning outcomes.

Theory module assessment methods:

- Multiple choice questionnaire
- Discussion (with notes)
 - These assessment modes will include Case studies i.e. what should you do in each situation?

Practical module assessment methods:

- Observation (observations to be supported by written notes and video/ photos)
- Verbal questions/discussion (recorded)
- Telematics (If available)

All evidence generated by the assessment process (such as completed test papers, audio and video recordings, photos) are to be retained in a secure location for review by the assessing body.

6.2. *KPIs for Measuring Programme Success

- **Fuel Consumption:**
 - **Average Fuel Savings:** Measure the reduction in fuel consumption (litres per 100 km or miles per gallon) before and after the training.
 - **Percentage Improvement:** Calculate the percentage reduction in fuel usage.
- **Driving Behaviour:**
 - **Incidents of Harsh Braking and Acceleration:** Monitor and reduce occurrences of harsh braking and acceleration.
 - **Steady Speed Maintenance:** Track how consistently drivers maintain a steady speed.
- **Vehicle Maintenance Costs:**
 - **Maintenance Cost Reduction:** Track the rate of vehicle maintenance costs due to better driving practices.
 - **Wear and Tear:** Monitor the rate of wear and tear on vehicle components (e.g., brakes, tires).
- **Driver Feedback:**
 - **Satisfaction Scores:** Collect and analyse driver satisfaction scores regarding the training program.
 - **Perceived Benefits:** Gather qualitative feedback on how drivers perceive the benefits of the training.

During the KPI reporting, the Approved HGV Eco Driving Master Trainer OR an Approved HGV Eco Driving Training Provider OR In-House Coach should also identify the correlation between these and other impactful measures such as:

- **CO2 Emissions:**
 - **Reduction in CO2 Emissions:** Express the saving in fuel use in terms of the resulting decrease in CO2.
 - **Percentage Reduction:** Express the reduction as a percentage to show environmental impact.
- **Safety Metrics:**
 - **Accident Rates:** Links between Eco-driving and changes in the number of accidents and near-misses generally.
 - **Driver Safety Scores:** Evaluate improvements in driver safety scores based on assessments and feedback.

7. Follow-Up Support

7.1. Weekly Feedback

- *Approved HGV Eco Driving Training Providers will provide companies and drivers with a resource pack / templates that facilitate ongoing performance monitoring,
- Drivers and companies are encouraged to use ongoing monitoring through apps and other methods
- In company continuous follow-up support is encouraged, with weekly feedback provided to drivers to maintain the benefits of coaching.
- Employers are encouraged to have a feedback mechanism in place prior to the start of the coaching programme.

8. *Certificates of Completion

8.1. Driver Certificates

- Certificates are issued to drivers as a form of recognition and appreciation for their efforts.
- A deliberate delay of two working weeks between the completion of coaching and certificate handover is preferred to allow performance results to be reviewed.

9.* Record Keeping & Reporting

9.1. Documentation

- Detailed records of each coaching session, including performance metrics and feedback, must be maintained.
- Records should be kept in a secure manner and be accessible for review for twenty four months following completion of the training.
- Approved HGV Eco Driving Training Providers will report quarterly on the number of drivers trained in the following:
 1. Drivers trained on theory and in-cab by the providers own approved HGV eco driving master trainer.
 2. Drivers trained on theory by the approved HGV eco driving provider, or master trainer and by in-house coach for in-cab training.
 3. In house coach trained by via the Approved HGV Eco Driving Training Provider

10. *Compliance and Assessment

10.1. Audit

- HGV Eco Driving Training Providers will submit programs to the assessing body for review. The submission must show how the provider has met the conditions outlined in this standard outlining the contact details/work experience of each Trainer
- Training providers will supply the assessing body with dates and locations where programs will run to facilitate unannounced assessment of training/visits. The Assessors will come from a designated SME Panel.
- Where the assessing body is satisfied that a program meets the standards outlined here, the training provider may use the logo of the Department of Transport HGV Eco-Driving Standard in the information and marketing materials for the program for twelve months following approval, unless otherwise withdrawn by notice in writing.

10.2. Assessment Process

- The assessing body will complete an unannounced assessment of 100% of all approved training providers in the calendar year.
- An unannounced assessment will consist of a subject matter expert attending training to observe the training methods and content used by the training provider. The subject matter expert may at their own discretion engage all attendees and administrative staff on and supporting the course and will compile a report to be shared with the assessing body and the
- Training provider. The auditor may, where vehicle construction allows, attend the in cab training as an observer and in any case examine records of in-cab training.
- The report will cover:
 - Confirmation of an Approved HGV Eco Driving Master Trainer OR an Approved HGV Eco Driving Training Provider OR In-House Coach for training delivery.
 - Suitability of venue – space, escape routes, interruptions, noise, toilets
 - Adherence to timetable
 - Performance of tutor – knowledge of topic, interaction with drivers
 - Compliance of training material
 - Overview of course delivery, including in-cab.

The report will have five action areas in its conclusions:

1. Commendations – activity the auditor observed that is of a high standard and the provider should be encouraged to develop and expand.
2. Recommendations – areas where the auditor believes that the provider may benefit from changes, but do not rise to the level of prohibiting continuing with the current program.
3. Conditions – activities or actions that must be rectified before the provider can enrol further students to the eco-driving program.
4. Breach – activities or actions that the provider must immediately rectify to continue delivery of courses using the Department of Transport HGV Eco Driving Standard

The subject matter expert will verbally summarise the report with the Approved Eco Driving Master Trainer or Approved Eco Driving Training Provider (if present) on the day of the assessment and a written copy of the full report will be provided to the Approved Eco Driving Training Provider within 6 weeks of the assessment.

In the case where the subject matter expert observes a “breach” this is an item for urgent response / mitigation and the Approved Eco Driving Training Provider will be informed immediately, the Approved Eco Driving Training Provider will be given written notification of this breach within 48 hours and a response with proposal for corrective action will be reviewed within 24 hours of receipt.

Where an Approved Eco Driving Training Provider has not satisfied the assessing body that the breach has been properly rectified the approval of the Department Transport HGV Eco-Driving will be withdrawn, and the provider will be required to remove all logos and brands from all advertising and training materials immediately.

The report will allow a right of reply for the provider where the provider may offer an explanation or observation on the SME report.

Approved Eco Driving Training Providers may appeal a decision of the Assessing body to the Department of Transport who will carry out a procedural review of the case. Department of Transport will have final decision.

Recommendations that have not been acted on may at the discretion of the subject matter expert, be amended to become conditions if observed on multiple (two or more) visits.

The assessing body will maintain at its discretion the right to undertake unannounced assessment of any approved training provider at any time regardless of the status of the assessment cycle. The assessing body may undertake this audit in response to, but not limited to, such issues as:

1. Student complaints about a provider's program.
2. Any concerns raised by Department of Transport.

Acknowledgements:

For contributions to the development of this standard, thanks are due to:

Freight Transport Association of Ireland

Road Safety Authority

Conor Molloy

Managing Director (AEMS)

Karl Lusted

CEO (Aptarus)

Carol Murphy

Training Centre Manager, Mayo Sligo Leitrim Education and Training Board

Jellinek Reinhard

Energy Agency Austria

Paul White

TM CPC Chief Examiner, DDLETB, Fleet Magazine.

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Distribution, Fleet and Compliance Manager, Spar BWG

Sean Murtagh

Director Fleet Magazine, TM CPC Corrector.

Paddy Sweeney

Managing Director REIL, on behalf of ENPROVA.

Appendix: Auditable Standards

1. Training Delivery

- a) Classroom-based instruction should be led by Approved HGV Eco Driving Master Trainer OR an Approved HGV Eco Driving Training Provider OR In-House Coach with appropriate knowledge/expertise, training to be a minimum of 2 hours.
- b) Trainers must have completed at a minimum Train the Trainer (level 6) or similar preparation.

2. Delivery Method

- a) Training content will include both theory and practical modules using face-to-face delivery.
- b) HGV Eco Driving theory training must be a minimum of 2 hours before any in-cab sessions.
- c) Training in-cab on-road driver observation, advice, and measurement.
- d) Results must be evidenced using the vehicle's trip computer or other appropriate data capture mechanism.

3. Learning Outcomes

- a) Drivers will understand HGV Eco driving principles and appreciate the environmental and economic benefits.
- b) Drivers will know practical techniques to improve fuel consumption and achieve measurable reductions in fuel usage.

4. Indicative Programme Content - Theory Module

- a) Introduction to Eco-Driving
- b) Driver Awareness and Augmented Reality Technologies
- c) Fuel Efficiency Techniques
- d) Vehicle Knowledge
- e) Driving Techniques
- f) Additional Eco-Driving Measures
- g) Review and Q&A

5. Practical Module

- a) First Drive: In-Vehicle Driver Evaluation over a minimum distance of 25Km.
- b) Fuel Efficiency and Safety Checks
- c) Second Drive: Driver Coaching over a minimum distance of 25Km.
- d) Final Evaluation

6. Performance Measurement

- a) Assessment Strategy: Multiple choice questionnaire, discussion, observation, verbal questions/discussion, telematics.
- b) KPIs for Measuring Programme Success: Fuel consumption, CO2 emissions, driving behaviour, safety metrics, vehicle maintenance costs, driver feedback.

7. Follow-Up Support

- a) Approved Training Providers will provide companies and drivers with a resource pack/template for ongoing performance monitoring.

8. Certificates of Completion

- a) Certificates are issued to drivers as recognition for their efforts.

9. Record Keeping

- a) Detailed records of each coaching session, including performance metrics and feedback, must be maintained for 24 months after the session.
- b) Appropriate records returned to the assessing body.

10. Compliance and Assessment

- a) Training Providers will submit programs to the assessing body for review.
- b) The assessing body will complete unannounced assessments of all approved training providers annually.